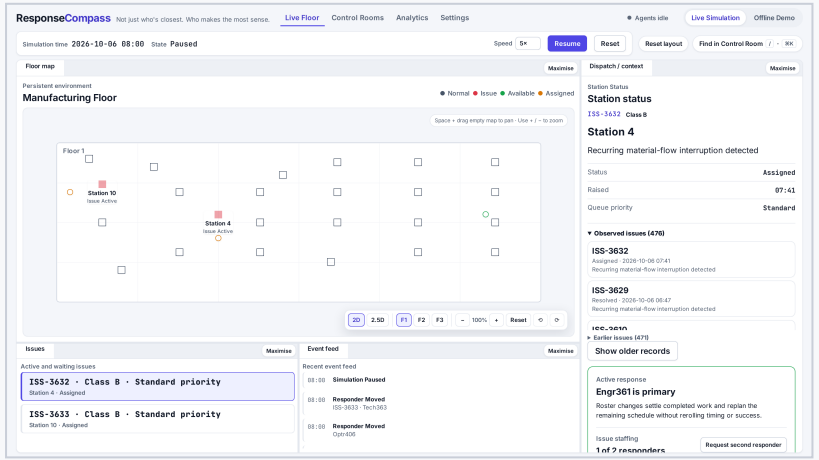
ResponseCompass

The right responder. The right fix.

Learned from every past repair.



Live Floor, simulated plant e8-aura-p1

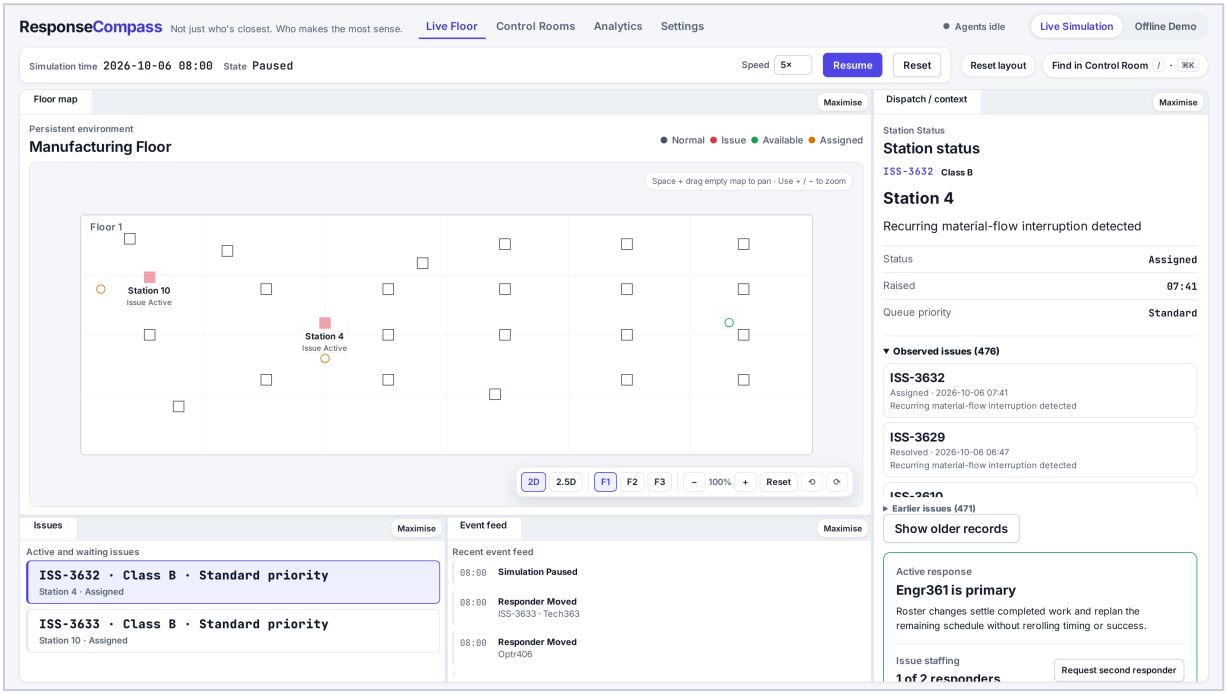
An AI dispatch agent for operations control rooms, built on AMD Token Factory.

The problem

- When equipment fails, **closest** or **free** may be an incomplete answer.
- The nearest technician may never have fixed this kind of fault.
- A poor match can mean downtime, a reopened issue or another trip.
- Who fixes what, and how, lives in people's heads and old tickets.

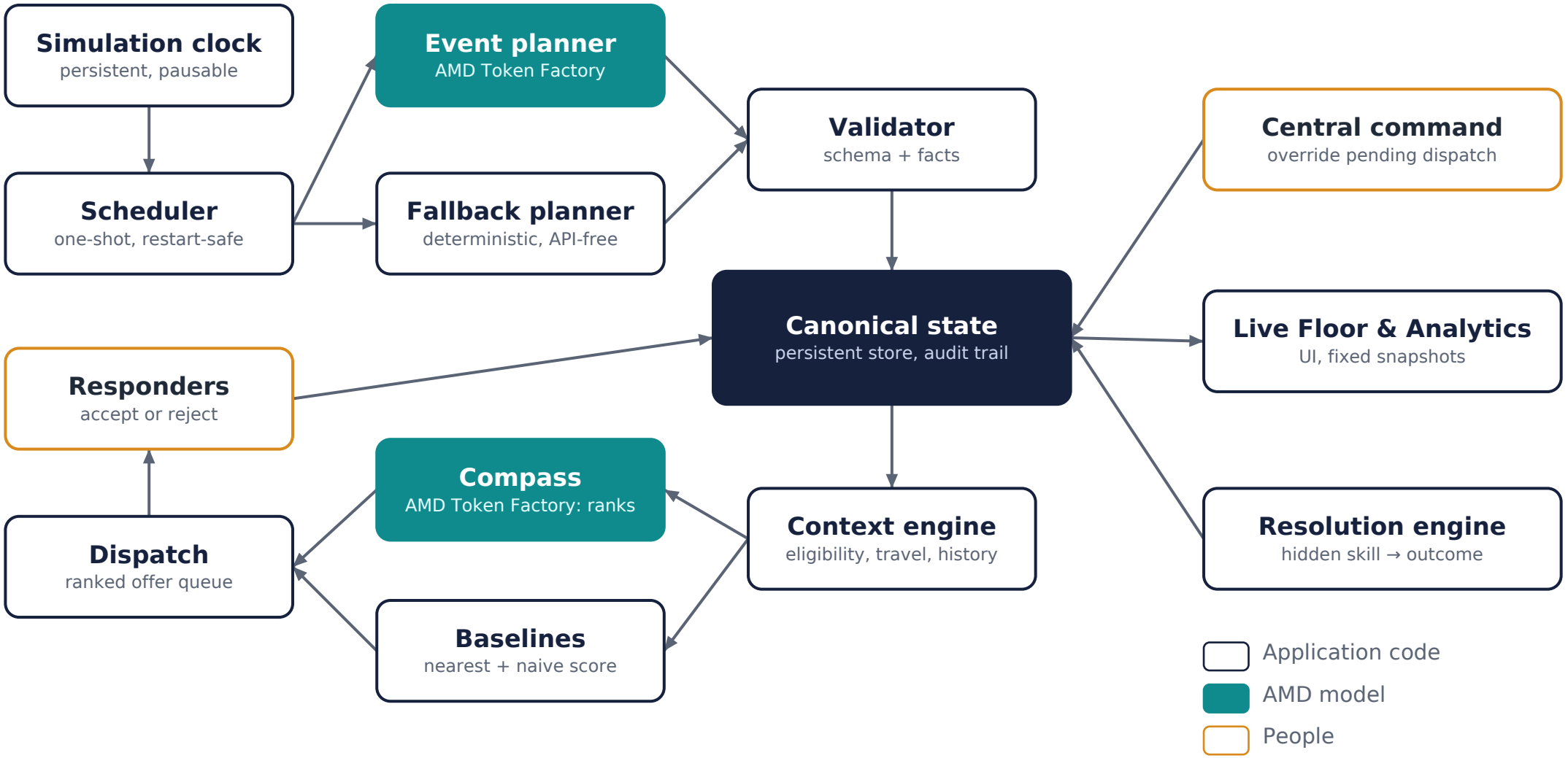
What ResponseCompass does

- For every new issue, Compass **ranks the eligible responders** and explains the tradeoffs.
- Offers go out automatically; responders accept or reject; complex work can get a second responder.
- Central command can override a pending dispatch; every step is audited.
- Each issue carries a **response action**; recommending fixes from past repairs is **planned**.



Code owns the facts. Models only rank.

Application code owns time, facts, eligibility, baselines and every mutation; models only propose.

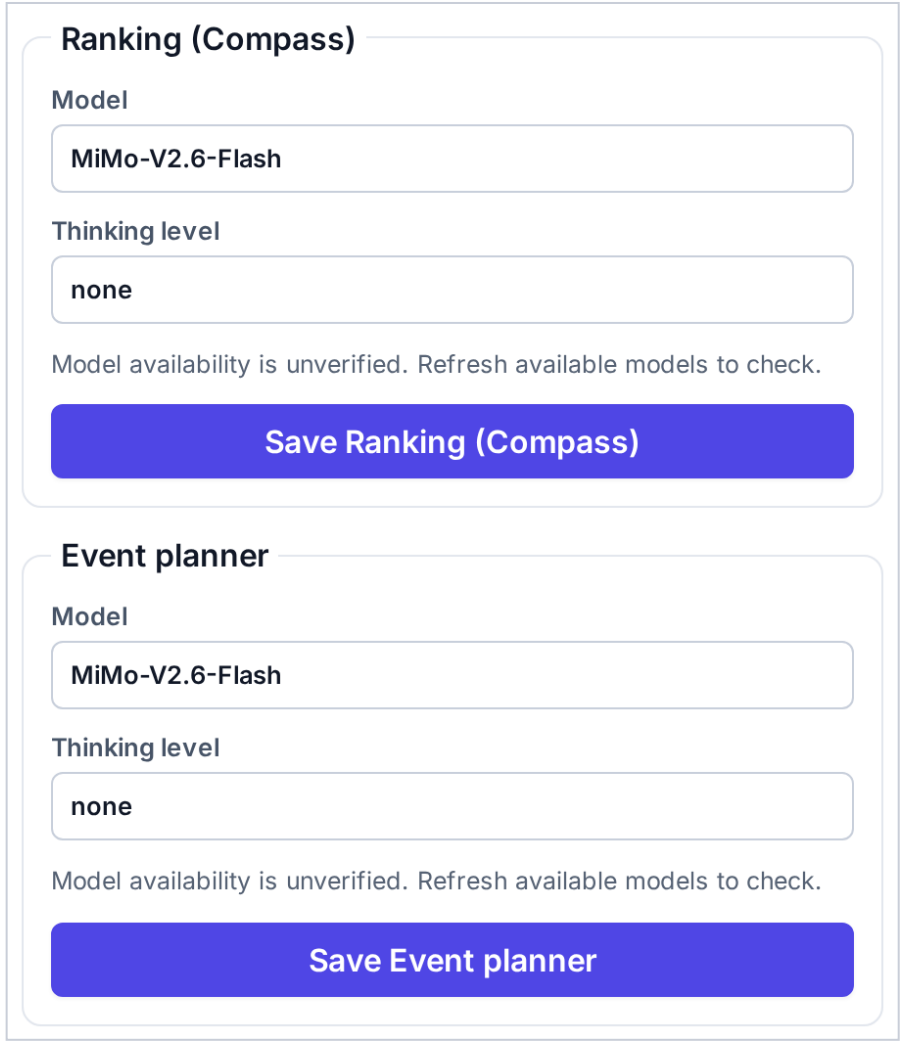


Honest by design

- Each responder's **true skill is hidden** simulation truth: no model, UI or analytics sees it.
- Outcomes are simulated from that hidden skill, independently of what the model claims.
- Two transparent baselines (nearest, naive score) run alongside; disagreements are flagged.
- Analytics show denominators, suppress thin samples and are not an employee leaderboard.
- Excluded from ranking: pay, seniority, demographics, fatigue, HR data.

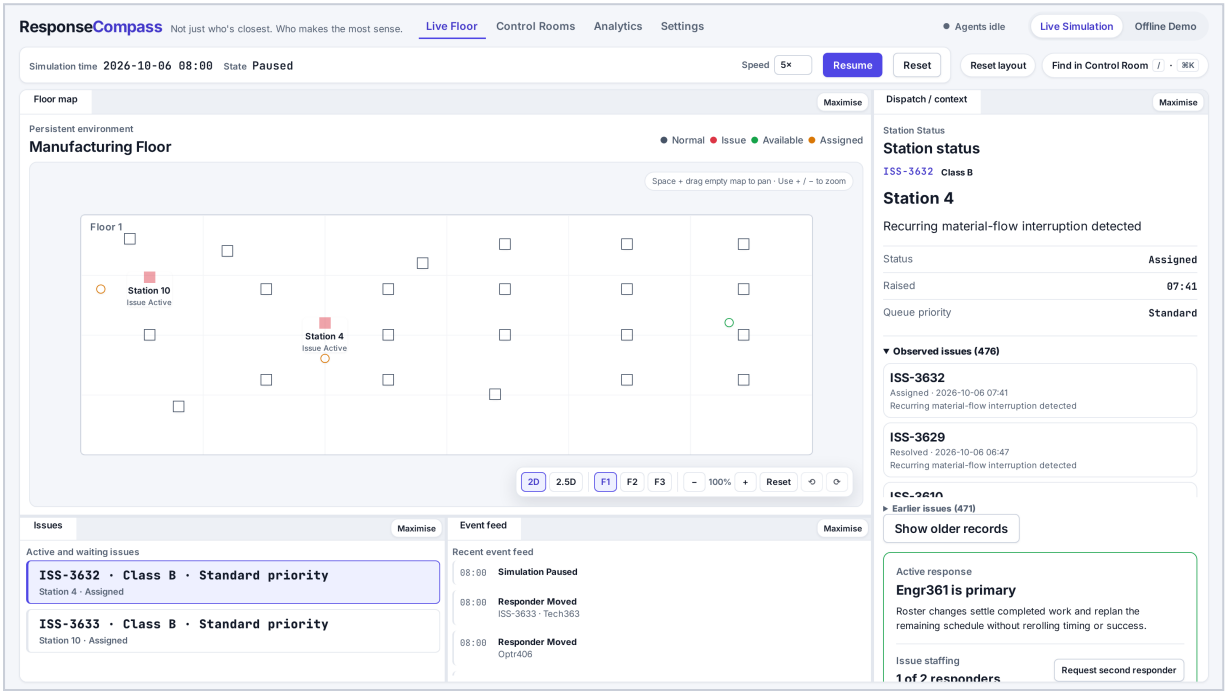
Built on AMD Token Factory

- OpenAI-compatible API, JSON output; our validators are the contract.
- Default model **MiMo-V2.6-Flash**; ranking and event planner can use different models.
- Local budget: 10 requests/min, 2 concurrent; bounded 429 retries; circuit breaker.
- Recorded MiMo check: **9/10** valid rankings, **10/10** valid event plans.
- Ranking deadline 90 s; model calls never hold the simulation clock.



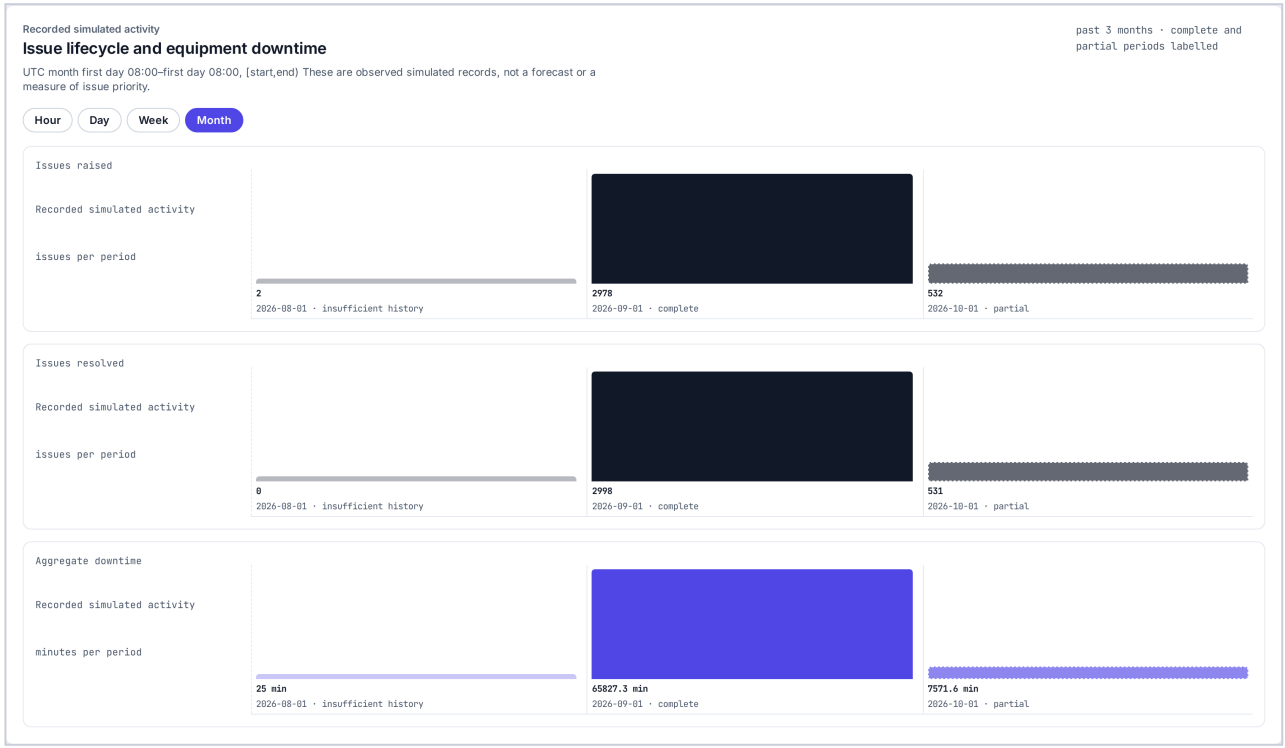
Live demo highlights

- Live Floor: three floors, 2D map and 2.5D overview, issues and responders in real time.
- Compass ranking with rationale; offer, accept, reject; help for complex work.
- Override and force-resolve; one-click audit trail per issue.
- Two independent control rooms (e8-aura-p1, e8-aura-p2).



A month of operations

- e8-aura-p1: **3,531 issues**, 1 September to 6 October (simulated).
- September: **2,978 raised / 2,998 resolved** events.
- Day, week and month views; Compass-vs-baseline records.
- Seven-day detail retention: live state served in **56 ms** in a local check.



Engineering quality

- Live and Offline are one app; Offline makes no API calls and is fully reproducible.
- **277** hermetic unit and integration tests and **51** Chromium end-to-end tests.
- Deterministic seeded simulation; restart-safe scheduler; atomic, crash-safe persistence.
- Runs on Node.js 22 or Docker; no external services besides the model API.

What's next

- **Recommended fixes:** rank response actions retrieved from comparable past resolutions.
- Learn from outcomes: feed accepted and successful fixes back into the evidence.
- Connect to real maintenance and ticketing systems in place of the simulated floor.
- Long-run storage: compact the remaining growing history for year-long runs.

The right responder. The right fix.